CS F364: Design & Analysis of Algorithms

Quiz 3 — June 9, 2026

Coverage: Lectures 1–12

> **Note**
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> **Instructions:**
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> - Time: 15 minutes.
> - Write your answers clearly on paper, scan, and upload to Google Classroom.
> - Show all steps for full credit.

Question: Median of Medians

The **Deterministic Select** algorithm (Median of Medians) finds the i^{th} smallest element in $O(n)$ time by:

1. Dividing the n elements into groups of **5**.
2. Finding the median of each group (by brute force, $O(1)$ per group).
3. Recursively finding the median x of these $\lceil n/5 \rceil$ medians.
4. Partitioning around x and recursing on the appropriate side.

The recurrence for groups of size **5** is:

$$T(n) \le T\left(\frac{n}{5}\right) + T\left(\frac{7n}{10}\right) + O(n)$$

(a) [2 marks] Give the recurrence $T_3(n)$ for the Deterministic Select algorithm when groups of size **3** are used instead of 5. What fraction of elements is guaranteed to be $\le$ the median-of-medians, and what fraction is guaranteed to be $\ge$ it?

(b) [2 marks] Give the recurrence $T_7(n)$ when groups of size **7** are used. What are the corresponding fractions?

(c) [1 mark] Solve $T_3(n)$ using the substitution method (guess $T_3(n) \le cn$ and show it fails, then state the actual asymptotic growth).

Solution

Part (a) — Groups of size 3

With group size 3, each group yields a median. There are $\lceil n/3 \rceil$ group medians.

- At least $\lceil \lceil n/3 \rceil / 2 \rceil \approx n/6$ medians are $\le$ the median-of-medians.
- Each such median has 2 elements $\le$ it in its group.
- So at least $2 \cdot \frac{n}{6} = \frac{n}{3}$ elements are $\le$ the median-of-medians.
- Similarly, $\frac{n}{3}$ elements are $\ge$ it.

Thus at most $\frac{2n}{3}$ elements are on either side in the worst case.

$$T_3(n) \le T\left(\frac{n}{3}\right) + T\left(\frac{2n}{3}\right) + O(n)$$

Part (b) — Groups of size 7

With group size 7, each group yields a median. There are $\lceil n/7 \rceil$ group medians.

- At least $\lceil \lceil n/7 \rceil / 2 \rceil \approx n/14$ medians are $\le$ the median-of-medians.
- Each such median has 3 elements $\le$ it in its group.
- So at least $4 \cdot \frac{n}{14} = \frac{2n}{7}$ elements are $\le$ the median-of-medians.
- Similarly, $\frac{2n}{7}$ elements are $\ge$ it.

Thus at most $\frac{5n}{7}$ elements are on either side in the worst case.

$$T_7(n) \le T\left(\frac{n}{7}\right) + T\left(\frac{5n}{7}\right) + O(n)$$

Part (c) — Solving $T_3(n)$

Guess: $T_3(n) \le cn$.

$$\begin{aligned} T_3(n) &\le T(n/3) + T(2n/3) + cn \\ &\le c \cdot \frac{n}{3} + c \cdot \frac{2n}{3} + cn \\ &= cn\left(\frac{1}{3} + \frac{2}{3} + 1\right) \\ &= cn(2) \\ &> cn \quad \text{(since } 2 > 1\text{)} \end{aligned}$$

The guess fails — the coefficients sum to > 1, so the recursion tree does **not** geometrically decay.

With $a/b = 1/3 + 2/3 = 1$, the actual growth is $T_3(n) = \Theta(n \log n)$ (see Quiz 2 for the reasoning: when the coefficients sum to 1, each level contributes $O(n)$ and there are $O(\log n)$ levels).

Summary

Group size	Reductions	Recurrence	Complexity
3	$2n/3$	$T(n/3) + T(2n/3) + O(n)$	$O(n \log n)$
5	$7n/10$	$T(n/5) + T(7n/10) + O(n)$	$O(n)$
7	$5n/7$	$T(n/7) + T(5n/7) + O(n)$	$O(n)$

Group sizes of 5 (or 7, 9, etc.) make the coefficients sum to < 1, giving geometric decay and $O(n)$ time.